MILITARY SPECIFICATION

STRAP, WRIST WATCH

1. SCOPE

1.1 This specification covers the material and fabrication requirements of a woven nylon strap intended for the attachment of a wrist watch to the wrist.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification of the extent specified herein.

SPECIFICATIONS

Federal

CCC-T-191	Textile Test Methods
PPP-B-636	Box, Fiberboard
PPP-T-76	Tape, Pressure Sensitive Adhesive, Paper, Water Resistant (for Carton Sealing)
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced

Military

MIL-P-116	Preservation, Methods of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel, Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

FSC 6645

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STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3.1 Preproduction samples. - Unless otherwise specified preproduction samples consisting of ten straps shall be furnished. The preproduction samples shall meet the requirements of this specification and shall be equal in all respects to the production units which the contractor proposes to furnish under this specification. The samples shall be manufactured utilizing the same methods, equipment and procedures as production units. Approval of the preproduction samples does not relieve the contractor of responsibility for compliance with all applicable provisions of this specification.

3.2 General. - Straps shall be furnished complete with buckle assembly.

3.3 Materials. - Materials shall be as specified herein. When materials are not specifically designated, they shall be entirely suitable for the intended purpose.

3.3.1 Yarn. - The yarn used in manufacture of the tape, from which the strap is fabricated, shall be continuous filament, bright, high tenacity nylon. It shall be a polyamide prepared from hexamethylene diamine and adipic acid, or its derivatives, and shall have a melting point of 482°F ± 10°F (250°C ± 6°F). The yarn shall be 210 denier, 34 filament.

3.3.2 Ply twist. - The filling yarn shall be 2 ply and shall have a minimum of 2-1/2 turns per inch. The warp yarn shall be single ply and shall have a minimum of 2-1/2 turns per inch. The plied yarns shall be twisted together (plied) in one operation. The final ply or yarn, whether single or 2 ply, shall be twisted so as to have a minimum of 2-1/2 turns per inch of "S" twist when taken from the webbing.

3.4 Weave. - The tape shall be composed of 2 ground warps, 1 ground warp weaving face and back, 1 ground warp weaving face to middle, 1 binder warp and 1 filling. The face and back warp shall weave plain with the picks that show on the face and back in the manner of plain tubular. The middle warp shall weave 2 up and 2 down with the face warp only. The binder warp shall weave 2 up and 2 down from face to back in the manner of a double shot.

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3.5 Tape. - The nylon tape shall conform to the following requirements:

Width, inch (± 1/32)	5/8
Weight (oz/yd) min.	.300
Thickness, inch (±.003)	.045
Warp ends, min. (total)	342
Ground, min.	214
Binder, min.	44
Face warp, min.	84
Picks per inch, min.	66
Ply warp, min.	Singles
Ply binder, min.	Singles
Ply face warp, min.	Singles
Ply filling, min.	2
Breaking strength, min.	800 lbs.

3.5.1 Nonfibrous materials. - Residual sizing, finishing, or other nonfibrous materials shall not exceed 2 percent of the dry weight of the tape.

3.5.2 Acidity. - The pH value of finished tape shall be within the range of 5 to 9.

3.6 Color. - Unless otherwise specified by the procuring activity, the color of the finished tape shall be olive drab and shall match No. X34087 of Federal Standard 595. The tape shall be dyed with acetate or acid type dyes. Metallized or chrome dyes shall not be used.

3.6.1 Colorfastness. - The dyed tape shall show colorfastness to light, after exposure for 20 standard fading hours, and laundering equal to or better than the standard shade sample when tested as specified in 4.2. When no standard sample is available for colorfastness the dyed tape shall have a colorfastness rating to light of good after exposure for 20 standard fading hours and good to laundering when tested as specified in 4.2.

3.7 Strap fabrication. - The strap shall be fabricated in accordance with Figure 1.

3.7.1 Buckle assembly. - The buckle shall be of a minimum size to accommodate the strap and shall have a tongue and pin of suitable size and shape. The buckle assembly shall be stainless steel material finished with a non-reflective surface and free of sharp edges.

3.8 Low temperature flexibility. - Straps shall show no evidence of breaking or splitting after being bent rapidly through an angle of 240 degrees around a 1/2 inch diameter metal mandrel at minus 20°F.

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3.9 High temperature-high humidity. - Straps shall show no evidence of damage affecting serviceability when subjected to 95% relative humidity at a temperature of plus 155°F.

3.10 Workmanship. - The finished tape shall be clean and evenly woven. The finished strap shall be finished in conformance with the highest industry practices of workmanship, general appearance and freedom from defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor responsibility. - The contractors inspection and quality efforts shall be such as to provide assurance that the supplies or services under the contract will conform to contract requirements and shall include procedures necessary to this end. When Military or Federal specifications are used to establish requirements in the contract the supplier shall be required to perform all examinations and tests called for by the contract requirements and specifications except those which are reserved for performance by the Government. Contractors are required to establish and maintain either inspection, and inspection system or a quality program, acceptable to the Government. The contractor is also required to maintain records of all inspection.

4.1.1 General provisions. - The quality assurance provisions of this specification and of other documents referenced herein form the basis for inspection to be performed by the supplier. Definitions of inspection terms not otherwise defined herein shall be as listed in Standard MIL-STD-109.

4.1.2 Preproduction samples. - Prior to production, the contractor shall submit the preproduction samples for inspection in accordance with 3.1. Should these samples fail to pass the prescribed tests, the contractor shall not submit any new samples until corrective action has been taken. Acceptance of production lots shall not commence until the preproduction samples have been accepted.

4.2 Control tests. - One sample of sufficient size for each control test shall be selected at random from each 10,000 yards of tape produced, or from each month's production, whichever comes first. Control samples shall have successfully passed all the tests specified in Table I prior to fabrication of straps.

4.2.1 Control test failure. - Should the control test sample fail to meet the specified test requirements, acceptance of the material will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved.

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TABLE I - CONTROL TESTS

CONTROL TEST	REQUIREMENT	TEST METHOD
Yarn	3.3.1	Certification
Melting point	3.3.1	4.5.1
Ply twist	3.3.2	*4054
Twist direction	3.3.2	*4050
Weave	3.4	Visual
Width	3.5	*5020
Thickness 1/	3.5	*5030
Thread count	3.5	*5050
Breaking strength	3.5	*5100
Non-fibrous material	3.5.1	4.5.2
Acidity	3.5.2	*2811
Color	3.6	Visual
Colorfastness to light 2/	3.6.1	*5660
Colorfastness to laundering 3/	3.6.1	*5614

*Test Methods of Specification CCC-T-191

1/ Except that a six ounce total load shall be applied, and the presser foot diameter shall be 3/8 inch.

2/ Exposure time shall be 20 standard fading hours.

3/ The test specimen shall consist of at least 5 linear inches of tape. The white color transfer cloth shall be the natural-color tape having the same dimensions as the tape specimen.

4.3 Inspection provisions.

4.3.1 Inspection lot. - Unless otherwise specified by the contracting officer, inspection lot size, formation, and presentation of lots shall be in accordance with "Submission of Product" as specified in Standard MIL-STD-105.

4.3.2 Examination and tests. - Examination and tests of completed straps shall be performed on a defect (individual characteristic) basis in accordance with Standard MIL-STD-105, and the inspection level and sampling plans specified in Table II titled "Classification of Defects". Examination and tests for Packaging, Packing and Marking shall be in accordance with Specification MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual non-conforming items.

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4.3.3 Disposition of non-conforming straps. - Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in Standard MIL-STD-105.

TABLE II - CLASSIFICATION OF DEFECTS

Use Level II of Table I with
Sampling Plan Table II-A of Standard MIL-STD-105

	REQUIREMENTS	TEST PROCEDURE
CRITICAL: NONE		
MAJOR: AQL 2.5%		
101. Strap fabrication	3.7	4.5.3
102. Heat seal	3.7	4.5.3.1
103. Buckle assembly	3.7.1	4.5.4
104. Low temperature flexibility	3.8	4.5.5
105. High temperature, high humidity	3.9	4.5.6
106. Workmanship	3.10	4.5.7
MINOR: NONE		

4.4 Inspection equipment. - Except as otherwise provided for by the contract, the contractor shall maintain all required measuring and testing equipment in accordance with the applicable requirements of Specification MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Government furnished inspection equipment. - Where the contract provides for Government furnished test equipment, care and maintenance of test equipment shall be in accordance with requirements of Specification MIL-I-45607.

4.5 Test methods and procedures.

4.5.1 Nylon melting point. - A test specimen of one milligram of nylon shall be used by cutting the fiber into lengths of 1/8 to 1/4 inch. The test shall be conducted by using an electrically heated stage having a circular depression large enough to insert a 19 M/M micro cover glass, and a powerstat, controlling the rate of heat input into the stage. The thermometer shall be a solid stem graduated from 20 to 300 degrees centigrade in one degree divisions (see 6.2). The specimen shall be placed in a small mound on a cover glass and covered with another cover glass. The two cover glasses shall be pressed together gently but firmly and placed in the circular depression on the stage. The temperature of the stage shall be rapidly raised to within 15 degrees centigrade of the expected melting point and thereafter at a rate of 3 to 4 degree centigrade per minute. Apply a slight pressure to the upper micro glass cover by pressing with a spatula. The temperature at which melting is first observed shall be the melting point as read to the nearest degree on the thermometer, and shall not exceed the limits of 3.3.1.

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4.5.2 Non-fibrous material. - The non-fibrous material, inclusive of the resin deposit, shall be determined on a specimen of approximately five grams of tape. The sample, after separation of the warp and filling yarns, shall be dried to constant weight in a weighting bottle at a temperature of 220 plus or minus 8 degrees Fahrenheit (104° ± 4.5°C). After a six hour extraction with methyl ethyl ketone in Soxhlet apparatus, the final weight of the extracted sample shall be obtained after constant weight has been obtained under the same drying conditions as before and shall not exceed the requirements of 3.5.1 when the following formula is used:

$$\text{Percent extractible matter} = \frac{\text{Loss in weight on extraction}}{\text{Original dry weight of sample}} \times 100$$

4.5.3 Strap fabrication. - Strap fabrication in accordance with Figure 1 shall be inspected by physically measuring the specified dimensions with a 12 inch scale. Searing of holes and slot shall be visually examined.

4.5.3.1 Heat seal test. - The heat seal of the strap and keeper shall be tested by pulling the seal apart. Examination of the separation shall indicate the presence of strap or keeper material. This test shall be performed with commercial equipment.

4.5.4 Buckle assembly. - The buckle assembly shall be visually inspected to the requirements of 3.7.1. Certification may be accepted for the material requirements, and a tactile examination shall be made to detect any sharp edges.

4.5.5 Low temperature flexibility. - Straps shall be conditioned for at least three hours in a cold chamber, maintained at the temperature specified in 3.8, within 2 degrees Fahrenheit. Following the exposure period, without moving the straps from the chamber, each strap shall be bent rapidly around the mandrel and examined for breaks or splits.

4.5.6 High temperature - high humidity. - The straps shall be placed in a test chamber that is capable of maintaining the specified temperature within two degrees Fahrenheit and the relative humidity within five percent. The straps shall be positioned in the test chamber with both surfaces exposed. The temperature and humidity shall be maintained at the values specified in 3.9 for a period of five hours. At the end of this period the straps shall be examined and shall show no physical change or damage which would affect any subsequent use.

4.5.7 Workmanship. - A visual examination shall be made to determine compliance with 3.10. The completed strap shall be of uniform quality and free from any defects.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. - Packaging shall be Level A or C as specified by the procuring activity (see 6.1).

5.1.1 Level A.

5.1.1.1 Cleaning. - Each strap shall be cleaned by process C-1 of Specification MIL-P-116.

5.1.1.2 Drying. - Each strap shall be dried in accordance with any procedure listed in Specification MIL-P-116 that shall not damage the strap.

5.1.1.3 Unit packaging. - The straps shall be packaged Method III of Specification MIL-P-116. The quantity of straps to be placed in the unit package shall be as specified by the procuring activity (see 6.1).

5.1.1.4 Intermediate packaging. - The quantity of unit packages to be placed in the intermediate container shall be as specified by the procuring activity (see 6.1). The intermediate container shall be a fiberboard box conforming to Class 2 W5c, of Specification PPP-B-636. Each fiberboard box shall be closed with pressure-sensitive tape as specified in the appendix to Specification PPP-B-636, except the tape shall conform to Specification PPP-T-76. The intermediate package may be used as the domestic shipping container when the total quantity to be shipped to one destination is equal to or less than the specified intermediate quantity.

5.1.2 Level C. - Cleaning, drying, and packaging shall be such that adequate protection is provided against corrosion, deterioration, and damage during direct shipment from the supply source to the first receiving activity for immediate use or for controlled humidity storage.

5.2 Packing. - Packing shall be Level A, B or C as specified by the procuring activity (see 6.1).

5.2.1 Level A. - Intermediate packages shall be packed in a fiberboard box conforming to Class 2, V3c, any style of Specification PPP-B-636. Closure of the fiberboard box shall be as specified in the appendix to the box specification. The fiberboard shipping container shall be reinforced with filament tape conforming to Type IV of Specification PPP-T-97. The tape shall be applied as specified in the tape specification.

5.2.2 Level B packing. - Intermediate packages shall be packed in a fiberboard box conforming to Class 1, Type I, any style of Specification PPP-B-636. Closure of the fiberboard box shall be as specified in the appendix to the box specification.

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5.2.3 Level C packing. - Packing shall be such as to afford protection against damage during direct shipment from the supply source to the first receiving activity in a manner to insure carrier acceptance. Containers shall be in accordance with carrier rules or regulations applicable to the mode of transportation.

5.3 Size of containers. - Interior packages and exterior containers shall be of a minimum size that will provide adequate protection to the contents.

5.4 Gross weight limitation. - The gross weight of fiberboard shipping containers conforming to Specification PPP-B-636 shall not exceed 70 pounds.

5.5 Marking. - Interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Selection of applicable levels of packaging, and packing
(c) Unit package quantity and intermediate package quantity
(d) Applicable stock number
(e) Color required (if other than that specified in 3.6)
(f) The Government laboratory where preproduction samples (see 3.1) shall be forwarded for testing.

6.2 Nylon melting point test apparatus. - An apparatus of the type described in 4.5.1 is manufactured by the "Fisher Scientific Co." and is called the "Fisher-Johns Melting Point Apparatus."

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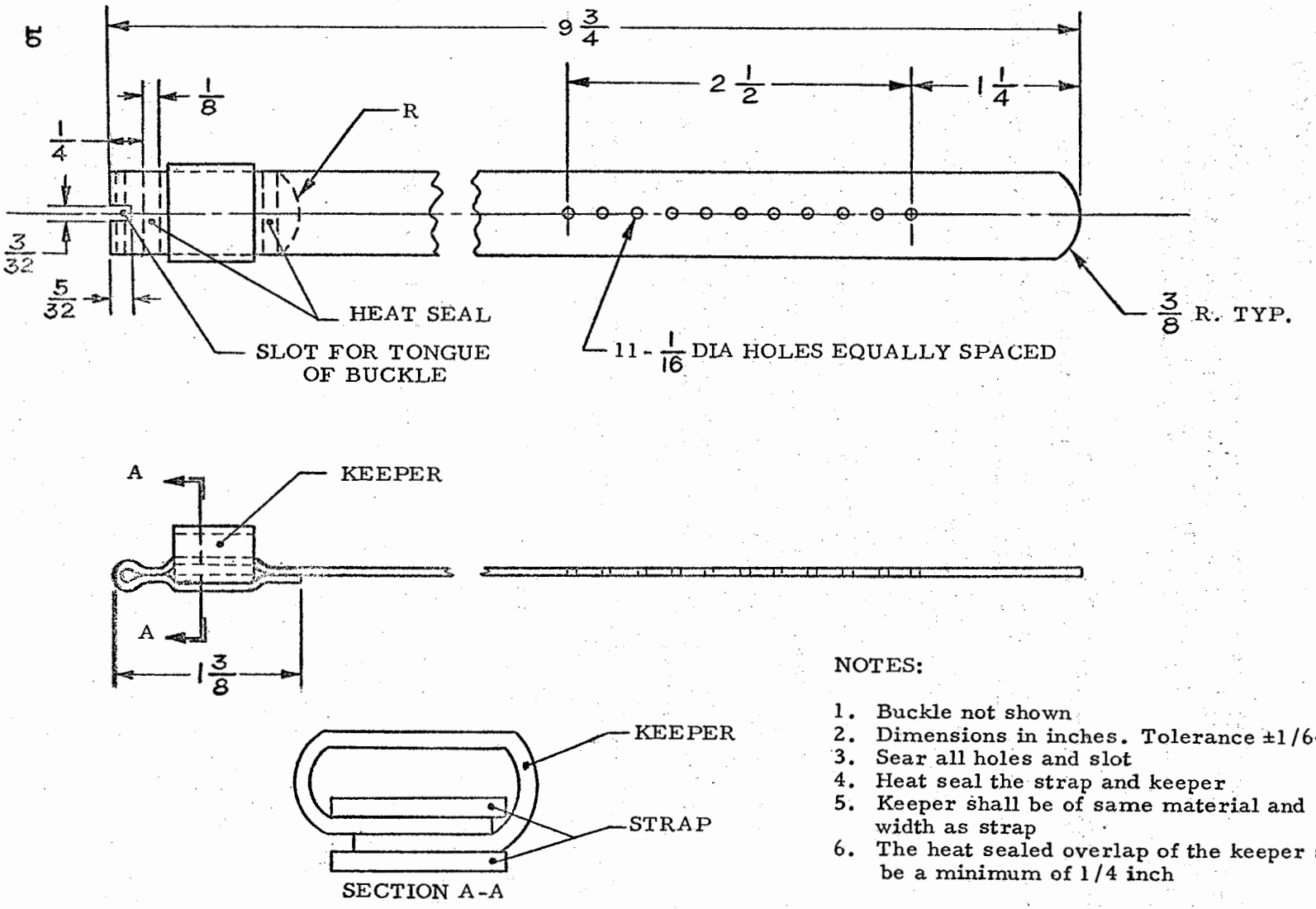
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NOTES:

1. Buckle not shown
2. Dimensions in inches. Tolerance ±1/64
3. Sear all holes and slot
4. Heat seal the strap and keeper
5. Keeper shall be of same material and width as strap
6. The heat sealed overlap of the keeper shall be a minimum of 1/4 inch

Figure 1

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